Codie Freeman

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Professional Summary

Final-year BSc Pharmaceutical Chemistry student (predicted 2:1, graduating June 2027) with 13 months' industry experience at a GMP contract research organisation. Currently applying machine learning to CYP450 inhibition prediction in the OpenADMET blind challenge.

Education

University of Reading | BSc Pharmaceutical Chemistry (Predicted 2:1) Sep 2021 - Jun 2027

Including foundation year and industrial placement year at Resolian

Experience

Resolian | Student Scientist Sandwich, Kent | Aug 2025 - Aug 2026

- **Project Delivery:** Executed analytical projects to tight sponsor deadlines, acting as lead analyst across multiple studies and working within teams on larger, multi-analyst programmes.
- **Method Development:** Designed an iGC-SEA flow-rate case study demonstrating up to 50% reduction in run time for larger probes, extending the laboratory's varied-flow-rate capability.
- **Technical Reporting:** Authored 10 multiple time-point stability reports to Ph. Eur. 2.9.19, taking each through QC and technical review and addressing reviewer comments before sponsor issue.
- **Data Analysis:** Recovered Dorris-Gray and Della Volpe surface energy parameters inaccessible through vendor software, processing raw instrument exports in Python and cross-checking outputs.
- **Process Improvement:** Identified an underutilised composite-imaging function, replacing a four-step manual composition with a single capture; created and delivered training resources to four analysts.
- **Data Integrity:** Maintained GMP laboratory notebooks, ADF records and audit trails to ALCOA++ standard, with sample logbooks and chain of custody kept current across all projects.

JD Wetherspoon | Shift Manager Reading, Berkshire | Jan 2022 - Jun 2025

- **Team Leadership:** Managed teams of up to 20 staff in a hospitality environment, overseeing onboarding, training and shift operations within a 90+ person team.
- **Operational Performance:** Improved average internal audit scores by 15%, contributing to the site's first full green audit in 10 years across cleanliness, stock, cash and people.
- **Regulatory Coordination:** Held personal licence, ensuring full compliance with licensing laws and leading a regulatory inspection that achieved a 5/5 rating.

Projects

Final Year Project: Machine Learning for CYP450 Inhibition Prediction Aug 2026 - Present

- Entrant in OpenADMET blind challenge using machine learning to predict inhibition of four cytochrome P450 drug-metabolising enzymes; designed and ran the full pipeline in Python.
- Designed a repeated cross-validation and statistical-testing framework to benchmark graph neural networks and molecular foundation models against tree-based baselines.

Surface Energy of Sucrose at Reduced Particle Sizes Dec 2025 - Mar 2026

- Characterised surface energy heterogeneity across three sucrose fractions produced by two processing routes, using iGC-SEA with Dorris-Gray and Peak COM analysis, alongside PXRD, SEM and BET surface area.
- Found that processing route governs surface energy character more strongly than particle size alone, with no direct relationship between BET specific surface area and surface energy, contrary to the expected trend.
- Presented at the Analytical Research Forum 2026 and subsequently adopted as a Resolian marketing resource.

Technical Skills

Materials Characterisation: iGC-SEA, DVS, PXRD, DSC, TGA, SEM, EDX, Light Microscopy

Analytical & Separation: HPLC-MS, GC, NMR, FTIR, UV-Vis

Quality & Compliance: GMP, ALCOA++, CAPA Processes, SOPs, COSHH, Risk Assessment, Data Integrity

Computational Analysis: Python (RDKit, Scikit-learn, Pandas, NumPy, Matplotlib), Jupyter, GitHub, Microsoft 365

Volunteering

STEM Learning UK | STEM Ambassador — Jul 2025 - Present

- Delivered a hands-on surfactant chemistry session to over 240 primary school pupils across two days at the East Kent Science Jamboree, Discovery Park.
- Engaged with school students on chemistry and pharmaceutical science careers through I'm a Scientist, an online scheme running live text-based Q&A between pupils and working scientists.

ELRIG | Red Shirt Volunteer — Oct 2026

- Supported delivery of Drug Discovery 2026 helping operations across two days for 3,000 attendees and sessions spanning AI-driven discovery to translational science.

Professional Development

Memberships: Royal Society of Chemistry, including the Chemical Information and Computer Applications Group (CICAG) and Joint Pharmaceutical Analysis Group (JPAG); Academy of Pharmaceutical Sciences.

Mentorships: University of Reading Thrive Mentorship and Royal Society of Chemistry mentoring scheme.

Certifications: University of Reading RED Award; CPD training in science communication and responsible AI.